ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: Halton Jersho. X

Roll No: 26CS207

Date:

Subject Name: Professional English Lab Integrated

Self Introduction: Written Script and One-Page Profile

Halton Jersho. X

Aspiring Software Developer and Technology Innovator

Links	**GitHub:** github.com/haltonjershoxps **LinkedIn:** linkedin.com/in/halton-jersho-x-8265943ab/ **Email:** haltonjershoxps@gmail.com
Summary	I am a first-year Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai. I am building my programming foundations, learning C, and exploring computational development and new technologies. My long-term aim is to grow into a software developer and technology innovator.
Introduction	Hi, I am Halton Jersho. X, a first-year CSE student in the 2026-2030 batch. I am learning programming with Python and C and building practical skills through coursework and coding practice. I want to keep improving, explore computational development, and discover new technologies.
Skills	Python; C (currently learning). Tools and platforms: ByteXL, HackerRank, LeetCode and ChatGPT. Operating system: Windows.
Project	**ATM Machine:** a basic program that prompts for a PIN and allows users to deposit and withdraw money.
Education	Computer Science and Engineering (CSE), first year. St. Joseph's College of Engineering, Chennai. Batch 2026-2030.
Awards	No awards, achievements or certifications reported.

Impressive Personality: Biographical Note

Lee Jae-yong (Jay Y. Lee), executive chairman of Samsung Electronics

Who He Is and His Field

Lee Jae-yong, widely known as Jay Y. Lee, is a South Korean business executive. Samsung Electronics appointed him executive chairman in October 2022. His career has moved through planning, customer relations, operations and senior leadership in a global technology company. Samsung's descriptions of his work are the company's own account of his contributions [1, 2].

Formative Years

Lee studied East Asian history at Seoul National University and later completed an MBA at Keio University. Samsung's 2016 executive director profile also records doctoral coursework at Harvard Business School [1]. His education combined study in the humanities with later business training.

Major Milestones

Lee joined Samsung Electronics in 1991. His responsibilities included strategic planning and customer relations, followed by operational leadership. In 2012 he became vice chairman, and in October 2022 the board appointed him executive chairman [1, 2]. This progression gives an example of how technical industries also depend on long-term planning, customer relationships and organizational leadership.

Leadership and Public Accountability

Lee's career has also been shaped by serious legal proceedings. In a bribery case, he was convicted and later received a two-year-and-six-month sentence at retrial; he was released on parole in 2021 and pardoned in 2022. A separate case about the 2015 merger of Samsung affiliates ended with the Supreme Court finalizing his acquittal in July 2025 [3, 4]. These are distinct cases and should not be conflated.

Professional Record

Lee's route through Samsung included strategic planning, customer-facing work and operations before his appointment as executive chairman. Samsung's own profile emphasizes these responsibilities; it is useful to read company profiles as an organization's description of an executive, rather than as an independent evaluation [1, 2].

Ethics and Resilience

The bribery conviction and its consequences raise questions about corporate governance, public trust and the responsibility that accompanies executive power. His later pardon enabled him to return to senior leadership, but did not erase that conviction. The separate merger case concluded with an acquittal in 2025. Keeping the legal histories distinct is important when describing his career [3, 4].

What I Take Away for My Engineering Path

I am a first-year CSE student building my programming foundation with Python and learning C. I want to develop practical computational skills and explore new technologies, including my idea of advancing virtual reality toward augmented reality or digital alternative reality. Lee's career reminds me that leadership in technology involves more than business success: it also calls for accountability, careful decisions and public trust.

Sources

[1] Samsung Electronics, 2016 Executive Director Nominee Profile.

[2] Samsung Global Newsroom, Jay Y. Lee appointed executive chairman, October 27, 2022.

[3] Associated Press, South Korea top court upholds acquittal in the 2015 merger case, July 17, 2025.

[4] Yonhap News Agency, Supreme Court finalizes acquittal in the 2015 merger case, July 17, 2025.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, “Developing Soft Skills”. That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person’s habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that “no chance” or a “splendid opportunity”? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey’s urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and “time stealers” such as waiting and travel can be turned into “time gifters.”

The lectures trace procrastination to Parkinson’s law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the “frog” (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying “yes” when you mean “no.”

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character (“when character is lost, everything is lost”). Professional ethics combine written codes with society’s moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi’s line that the earth provides enough for everyone’s need but not everyone’s greed.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides 1 and 2)

Strategic Career Roadmap: SWOC Analysis

Building toward new technologies

Candidate: Halton Jersho. X | First-year CSE | Batch 2026-2030

A learning-first plan grounded in current skills and goals

Executive Summary & Career Vision

1 Vision

Explore and discover new technologies.

2 Objective

Build practical ability in computational development.

3 Framework

Learn foundations, deepen skills, then apply them in projects.

A practical path: build confidence through fundamentals, practice and projects.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides 3 and 4)

SWOC: Personal Starting Point

Strengths and areas for growth

S Strengths

- Adaptability
- Multitasking
- Responsibility
- Python foundations

W Growth Areas

- Build consistency
- Practice steady habits
- Focus on one task before switching

Based on Halton's stated strengths and development areas.

SWOC: Opportunities and Challenges

Conditions that can shape the learning path

O Opportunities

- NPTEL courses
- Hackathons and technical clubs
- Student cloud credits
- Open-source projects

C Challenges

- Managing workload
- Competition
- Fast-changing tools
- Cost of learning resources

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides 5 and 6)

Strategic Actions

Turn self-awareness into steady progress

1 Consistency

Set a manageable routine and finish small tasks regularly.

2 Focus

Work on one task at a time; use short checklists.

3 Practice

Strengthen C and apply learning in small projects.

Use coding practice, coursework and project work to reinforce each new skill.

Four-Year Learning Focus

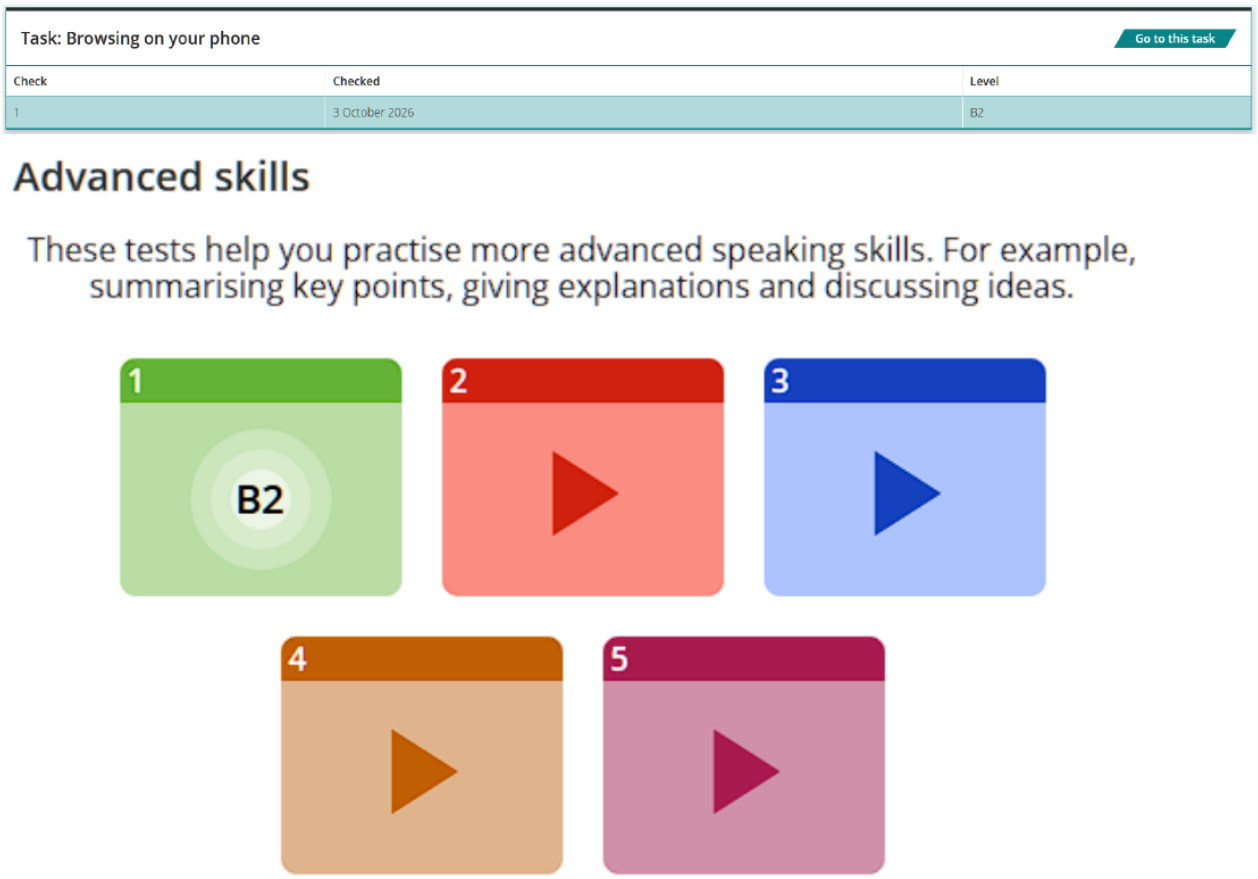
A flexible progression; focus areas can evolve with experience

Year	Focus	Direction
Year 1	Foundation	Build programming fundamentals and strengthen C skills.
Year 2	Development	Learn additional programming languages and development skills.
Year 3	Exploration	Explore advanced areas of computational development.
Year 4	Career preparation	Apply technical knowledge through projects and prepare for future opportunities.

Planning targets from Halton's fact sheet; no specific deliverables or credentials were reported.

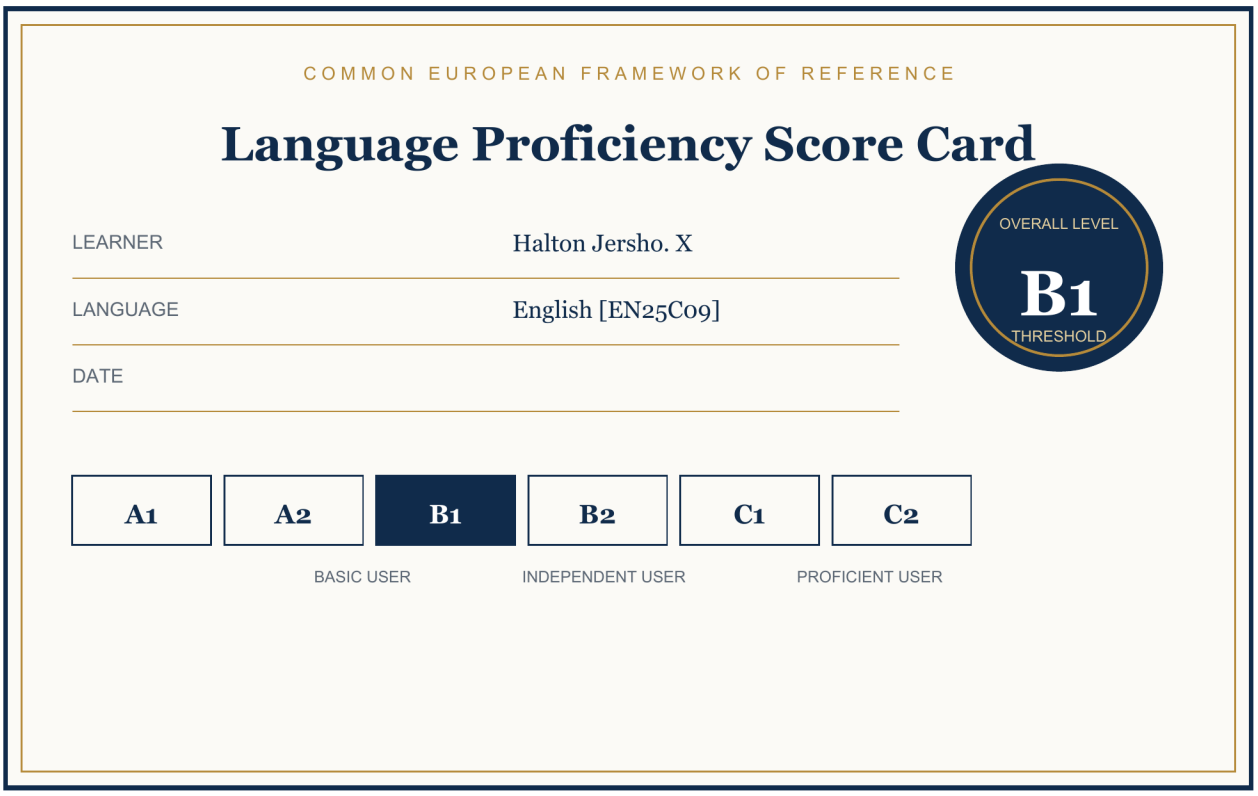
Write and Improve / Speak and Improve

Speak and Improve: score sheet (Task: Browsing on your phone)



Write and Improve / Speak and Improve

CEFR score card



My CEFR level is B1 (Intermediate).

B1 at a glance

Familiar topics

Can handle familiar topics

Everyday situations

Can handle everyday situations

Experiences and plans

Can describe experiences and plans

Opinions

Can give brief reasons for opinions

Overall level reported: B1. The assessment date and separate writing/speaking scores were not provided.

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must
5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

    (b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for every account.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitive information.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or the National Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Bharatanatyam: The Dance That Carries Tamil Nadu's Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Halton Jersho. X
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Halton Jersho. X
(Register No.: 26CS207

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]

Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Halton Jersho. X

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Halton Jersho. X
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Halton Jersho. X
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Halton Jersho. X
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Halton Jersho. X

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Halton Jersho. X
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Halton Jersho. X
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Halton Jersho. X
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹[amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Halton Jersho. X
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Halton Jersho. X

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Halton Jersho. X

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Halton Jersho. X

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Project Proposal: Ideation

Advanced Reality Experience: exploring a path from VR toward AR or DAR

Halton Jersho. X | Early-stage concept | Based on the project canvas

1. Overview

This idea explores how virtual reality could develop toward augmented reality (AR) or digital alternative reality (DAR), bringing digital spatial content into a real-world setting. The concept is intended to help learners connect a virtual 3D view with a physical workspace. It is an early proposal: a specific task, user need, device, budget and prototype have not yet been selected or validated.

2. Working Hypothesis

Engineering learners may find it difficult to connect a virtual 3D representation with the real workspace where an object or process is used. A small AR experience could make spatial information easier to inspect alongside the physical setting. This is a hypothesis to check with students or faculty, not a confirmed problem.

3. Proposed Scope

The target is a small prototype for one validated campus or engineering task. The next step is to identify that task and confirm the need with potential users before selecting technology or building anything.

4. Problem Definition

Working hypothesis: learners may have trouble relating a virtual 3D view to a real workspace. The exact context, task and user pain point are still unknown. The project should first find out whether this challenge occurs in a specific learning activity and whether overlaying digital content would help.

5. Objective

Explore a path from VR toward AR or DAR through one small, testable experience. The intended value is to make selected spatial content easier to view alongside a physical setting. The concept will proceed only after a task and user need are confirmed.

6. Non-goals at This Stage

- Claiming that an existing prototype is complete.
- Selecting a device, AR platform, SDK or implementation language before access is confirmed.
- Promising learning gains, adoption or commercial impact before testing.
- Designing a general-purpose system for every engineering task.

7. Potential Users

Possible users are engineering students or instructors working with 3D concepts. These groups are candidates only; they have not yet been interviewed or asked to confirm the problem.

8. Intended Value

If a real learning task is identified, showing digital spatial content beside the physical workspace could help a learner inspect, compare or adjust a 3D concept. Whether this makes the task clearer or easier is not known yet and must be evaluated with users.

9. Design Principles to Test

- Keep the first prototype limited to one agreed task.
- Make digital placement and alignment easy to inspect.
- Test stability and ease of use with the intended users.
- Explain clearly when a view is approximate or needs adjustment.

Impact Status

Learning benefit, adoption and social or economic effects have not been measured. The project should report evidence from its first user tests rather than assume outcomes.

Project Proposal: Ideation

Candidate experience, architecture and scope

10. Concept Workflow

1. Choose a task. Agree on one campus or engineering activity where students use a 3D concept.

2. Capture the scene. A candidate device camera and motion sensors provide input; hardware access needs confirmation.

3. Track the view. Visual feature or plane tracking, combined with sensor data, could estimate the device view and anchor points.

4. Place digital content. A selected AR API could position a chosen 3D item in the real scene for inspection or adjustment.

5. Test the result. Check alignment, stability and ease of use during the agreed task.

Open Design Question

The content format, interaction method and exact task will be selected only after the need and available devices are confirmed.

11. Proposed Technical Architecture

The canvas suggests a candidate flow rather than a selected stack:

Input: device camera and motion-sensor data.
Tracking: visual feature or plane tracking combined with sensor data to estimate the view and anchor points.
Placement: an AR API places selected digital content in the real scene.
Evaluation: a short task checks alignment, stability and ease of use.

12. Current Starting Point

Halton reports Python, C (currently learning), Windows, ByteXL, HackerRank, LeetCode and ChatGPT. This describes his current learning profile; it does not establish that these tools are suitable for building an AR prototype.

Technology Selection

No AR API, SDK, tracking method, development language or 3D content tool has been selected. The options require research and testing after device access is confirmed.

13. First Prototype Scope

A first prototype should address one validated campus task and one agreed user group. The minimum scope is to place or view a small amount of digital spatial content in the physical scene and evaluate how well it supports the task.

14. Dependencies to Confirm

- A device with a camera and motion sensors.
- Access to a suitable AR API or SDK.
- A source or method for creating/selecting 3D content.
- Permission and access to test in the chosen setting.
- A budget and deployment context.

15. Constraints

Device access, tools, budget and deployment setting are not yet known. The technical design should stay small until these constraints and the target task are confirmed.

Out of Scope for the Current Idea

No broad deployment plan, finished product, validated use case or tested implementation has been supplied.

Project Proposal: Ideation

Validation, risks and next steps

16. Before Building

- Choose one engineering or campus task where spatial information matters.
- Speak with students or faculty who perform that task.
- Confirm whether connecting the 3D view to the physical workspace is a real difficulty.
- Check available devices, budget and permission to test.

17. Prototype Evaluation

If the need is confirmed and a device is available, build only the smallest experience needed to test the idea. Observe whether the overlay stays aligned, remains stable during the task and feels easy for intended users to understand and adjust.

18. Success Measures

The project canvas names alignment, stability and ease of use as checks. Specific thresholds and a test method still need to be chosen with users before evaluation.

Evidence to Record

Document the task, participants' feedback, device setup, observed problems and proposed changes. Do not report learning improvement unless a suitable comparison is completed.

19. Key Risks

Unconfirmed user need: the project may not address a real difficulty.
Device access: suitable camera and motion sensors may not be available.
Tracking quality: overlays may drift or feel unstable.
Scope growth: the broad VR-to-AR idea can expand beyond a first-year project.
Cost: tools or hardware may not fit the available budget.

20. Mitigation

Validate one task with users before choosing tools. Confirm device access and budget early. Keep the first prototype small, test alignment and stability in the intended setting, and record limitations. Stop or revise the idea if the need or hardware support is not confirmed.

Status

These are planning risks for an early concept. No risk has been tested against a working prototype.

21. Proposed Sequence

Step 1: Select one campus or engineering task and check the need with students or faculty.
Step 2: Confirm device access, budget and test setting.
Step 3: Compare possible AR APIs and tracking options for that device.
Step 4: Create a small prototype for the chosen task.
Step 5: Evaluate alignment, stability and ease of use, then decide whether to continue.

22. Decisions Still Open

The specific user task, confirmed problem, hardware, API/SDK, content source, budget, test plan and timeline remain undecided. No prototype or deployment plan has been reported.

23. Conclusion

The project is a promising direction for learning about spatial computing, but it is still an idea. Its next useful result is evidence that a particular task and user need justify a small, feasible prototype.

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

Halton Jersho. X

Aspiring Software Developer and Technology Innovator

CSE Student (Batch 2026-2030) | Python and C | github.com/haltonjershoxps

About

I am a first-year Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai. I am building my programming foundation with Python and learning C. I want to develop practical skills in computational development, keep learning, and explore new technologies.

Focus Areas

Programming foundations; computational development; learning C; coding practice; emerging technologies.

Featured Project

ATM Machine
A basic program that prompts for a PIN and allows users to deposit and withdraw money.
Current tools reported: Python and C learning. No project repository link was supplied.

Education

Computer Science and Engineering (CSE), first year. St. Joseph's College of Engineering, Chennai. Batch 2026-2030.

Skills

Programming: Python; C (currently learning). Operating system: Windows. Tools and platforms: ByteXL, HackerRank, LeetCode and ChatGPT.

LinkedIn and GitHub Profiles

Career roadmap and development plan

Career Direction

Explore and discover new technologies while building practical ability in computational development. The four-year outline below reflects a flexible learning direction, not a list of completed achievements or guaranteed deliverables.

Year	Focus	Direction
Year 1 (2026-27)	Foundation	Build programming fundamentals and strengthen C skills.
Year 2 (2027-28)	Development	Learn additional programming languages and development skills.
Year 3 (2028-29)	Exploration	Explore advanced areas of computational development.
Year 4 (2029-30)	Career preparation	Apply technical knowledge through projects and prepare for future opportunities.

Certifications

No completed or planned certifications were reported.

Growth Areas

Build consistency and stronger study habits. Practice focusing on one task at a time when multitasking makes concentration harder.

Opportunities and Challenges

Opportunities: NPTEL, hackathons, student cloud credits, open-source projects and technical clubs. Challenges: workload, competition, fast-changing tools and cost.

LinkedIn and GitHub Profiles

Part B: GitHub profile README

Hi there, I'm Halton Jersho. X

CSE student | Aspiring software developer and technology innovator

Connect

GitHub: github.com/haltonjershoxps
LinkedIn: linkedin.com/in/halton-jersho-x-8265943ab/
Email: haltonjershoxps@gmail.com

About Me

- I am a first-year CSE student at St. Joseph's College of Engineering, Chennai.
- I am building programming foundations with Python and learning C.
- I use coding practice and coursework to improve my skills.
- My long-term goal is to explore and discover new technologies.

Featured Project: ATM Machine

A basic program that prompts for a PIN and allows users to deposit and withdraw money. A repository URL was not provided.

Project Idea

Explore advancing VR toward AR (augmented reality) or DAR (digital alternative reality), bringing digital spatial content into a real-world setting. The idea is at an early stage; a specific use case still needs validation.

Current Toolkit

Python; C (currently learning); Windows; ByteXL; HackerRank; LeetCode; ChatGPT.